

SIGNEX — SIGNAL EXTRACTION & ANALYSIS ENGINE

NTRO ID26147 CLASSIFIED INTELLIGENCE DOSSIER

CLASSIFICATION: SECRET // AIR-GAPPED INTERNAL // REF: 06697ad0e5e65ace

INTERCEPT METADATA & CHAIN OF CUSTODY

Source Filename : real_air_traffic_control_broadcast.iq
File SHA-256 Digest : 06697ad0e5e65ace170b17dff0a88b4ac68c6add6b05296c33d6831faa988edd
Payload File Size : 524,288 bytes (512.0 KB)
Analysis Timestamp : 2026-09-19T08:30:34Z
Operator ID : SIGINT-AIRGAP-OPERATOR
AMC Model Engine : amc_resnet18_v1.0 (Deep 1D ResNet-18 + Ensemble)

EXTRACTED RF SIGNAL PARAMETERS

Parameter Description	Estimated Value	Extraction Metric / Method
Sampling Frequency (F_s)	2.0000 MSps	Baseband Nyquist Rate
Center Frequency (F_c)	+6.243 kHz	Spectral Centroid Offset
Estimated Baud Rate	250.00 kBaud	Cyclostationary Wavelet Energy
3-dB Bandwidth	976.6 Hz	Half-Power Frequency Span
Occupied Bandwidth (99%)	1740.72 kHz	Cumulative PSD 99% Energy
Estimated SNR	8.19 dB	Fourth-Order Cumulant (C_{42})
Peak-to-Average Power (PAPR)	3.65 dB	Temporal Envelope Crest Ratio
Carrier Frequency Offset (CFO)	+0.00 Hz	Non-Linear M-th Power Loop
Spectral Flatness (Entropy)	0.1396	Wiener Entropy Index (0.0 – 1.0)

PREDICTED MODULATION : WBFM

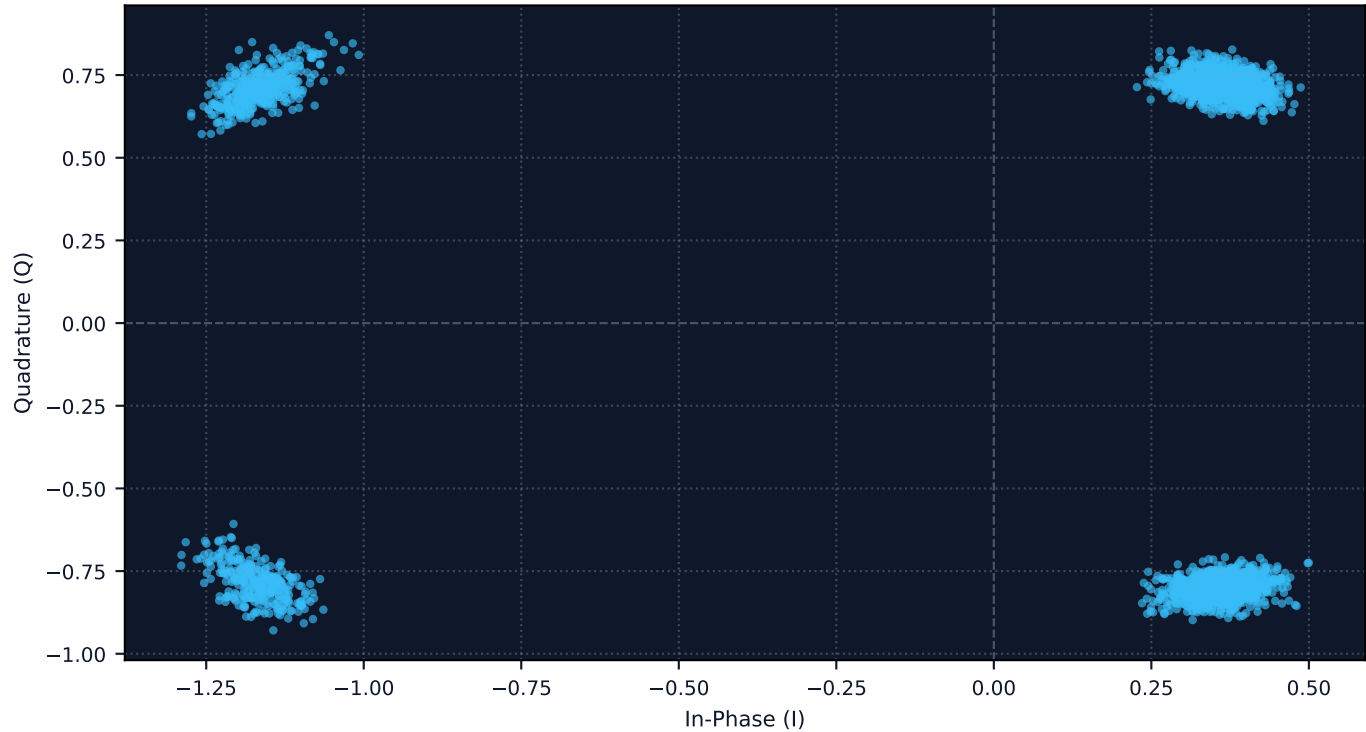
CLASSIFIER CONFIDENCE : 81.14% (ResNet-18 Softmax Ensemble)

INTELLIGENCE STATUS : CONFIRMED: HIGH CERTAINTY MODULATION CLASSIFICATION

Demodulation Scheme: QPSK | Recovered Bits: 16,384 | Estimated EVM: 10.00% | CRC Validated Frames: 0

SIGNAL DISSECTION & DEMODULATION INTELLIGENCE // [WBFM]

Demodulated Baseband Constellation Diagram — [WBFM]



SYNCHRONIZED TELEMETRY FRAMES & PROTOCOL DISSECTION

Frame #	Sync Index	Tx ID	Seq No.	Payload	CRC Status	FEC Status
#1	0	0x0000	0	2041 B	CRC MISMATCH	PARTIAL_RECOVER
#2	254	0x0000	1	2009 B	CRC MISMATCH	PARTIAL_RECOVER
#3	317	0x0000	2	2001 B	CRC MISMATCH	PARTIAL_RECOVER
#4	501	0x0000	3	1978 B	CRC MISMATCH	PARTIAL_RECOVER
#5	544	0x0000	4	1973 B	CRC MISMATCH	PARTIAL_RECOVER

DECODED BITSTREAM HEX / ASCII PREVIEW (FIRST 64 BYTES)

0000 A98909240A89EAE8AA440A89E408C989 [FRAME 1 PAYLOAD]